

PURVA TAGDE
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BATCH B1

FINAL TASK

```
24_9_25.html  <> pract5_final_task.html X  <> 10_9_25.html  <> pract3_final_presentation.html  # grading.c
JavaScript > <> pract5_final_task.html > script > cart.forEach() callback
1  <script>
2      let cart = [
3          { id: 1, name: "Pen", price: 20, qty: 2, category: "stationery" },
4          { id: 2, name: "Mug", price: 150, qty: 1, category: "kitchen" },
5          { id: 3, name: "Notebook", price: 80, qty: 3, category: "stationery" }
6      ];
7
8      let discountedItemTotals = cart.map(item => {
9          let subtotal = item.price * item.qty;
10         if (item.qty >= 3) {
11             subtotal -= subtotal * 0.05;
12         }
13         return subtotal;
14     });
15
16     let stationeryItems = cart.filter(item => item.category === "stationery");
17     let discountedStationeryTotal = stationeryItems.reduce((sum, item) => {
18         let total = item.price * item.qty;
19         if (item.qty >= 3) total -= total * 0.05;
20         return sum + total;
21     }, 0);
22
23     let stationeryDiscount = 0;
24     if (discountedStationeryTotal > 200) {
25         stationeryDiscount = discountedStationeryTotal * 0.10;
26     }
27
28     let subtotal = cart.reduce((sum, item) => sum + (item.price * item.qty), 0);
29     let totalAfterItemDiscounts = discountedItemTotals.reduce((sum, val) => sum + val, 0);
30     let itemDiscount = subtotal - totalAfterItemDiscounts;
31     let finalTotal = totalAfterItemDiscounts - stationeryDiscount;
32
33     cart.forEach(item => {
34         let total = item.price * item.qty;
35         if (item.qty >= 3) {
36             console.log(`Item: ${item.name} (x${item.qty}) = ${total} + discount applied`);
```

```

33 cart.forEach(item => {
34   let total = item.price * item.qty;
35   if (item.qty >= 3) {
36     console.log(`Item: ${item.name} (x${item.qty}) = ${total} + discount applied`);
37   } else {
38     console.log(`Item: ${item.name} (x${item.qty}) = ${total}`);
39   }
40 });
41
42 console.log("-----");
43 console.log("Subtotal:", subtotal);
44 console.log("Item Discounts:", itemDiscount.toFixed(2));
45 console.log("Stationery Discount:", stationeryDiscount.toFixed(2));
46 console.log("Final Total:", finalTotal.toFixed(2));
47 </script>

```

Elements Console Sources Network >> 1

top Filter Default levels 1 Issue: 1

Item: Pen (x2) = 40	pract5_final_task.html:38
Item: Mug (x1) = 150	pract5_final_task.html:38
Item: Notebook (x3) = 240 + discount applied	pract5_final_task.html:36
-----	pract5_final_task.html:42
Subtotal: 430	pract5_final_task.html:43
Item Discounts: 12.00	pract5_final_task.html:44
Stationery Discount: 26.80	pract5_final_task.html:45
Final Total: 391.20	pract5_final_task.html:46

>

SOLVE THE FOLLOWING:

Que.1.

Which function (map, filter, reduce, forEach) was hardest to apply? Why?

=>reduce() – because it has to accumulate totals while applying conditional discounts.

- **map() just transforms items.**
- **filter() selects items by category.**
- **forEach() is only for printing, doesn't return values.**

Example:

```
let discountedStationeryTotal = stationeryItems.reduce((sum, item) => {  
  let total = item.price * item.qty;  
  if(item.qty >= 3) total -= total * 0.05;  
  return sum + total;  
}, 0);
```

Que.2.

How could this program be extended (e.g., tax, coupons, shipping)?

=>

- **Tax: Add a percentage of final total.**
- **Coupons: Apply flat or percentage discounts.**
- **Shipping: Add charges based on total amount.**
- **More category discounts: Different rules for electronics, kitchen, etc.**